

Artificial Intelligence

Lecture 05 – Forward & Backward Chaining

Dr. Rajiv Misra, Professor
Dept. of Computer Science & Engg.
Indian Institute of Technology, Patna
rajivm@iitp.ac.in



Inference – Horn Clause

Horn Form

- **Horn Clause** (at most one literal is Positive or non-Negated)
- For example:
 - $(\neg P \vee \neg Q \vee V)$ is a Horn Clause.
 - so is $(\neg Q \vee \neg V)$
 - but $(\neg P \vee Q \vee V)$ is not
- **Definite Clauses**: exactly one literal is positive.
- KB as a conjunction of Horn clauses
- Horn clauses can be re-written as implications
 - proposition symbol (fact) or
 - conjunction of symbols (body or premise) $\rightarrow$ (conclusion or head)
 - Example: $(\neg C \vee \neg B \vee A)$ becomes $(C \wedge B \rightarrow A)$
- Modus ponens for Horn KB:
$$\frac{\alpha_1 \dots \alpha_n, \alpha_1 \wedge \dots \wedge \alpha_n \Rightarrow \beta}{\beta}$$

Inference – Forward Chaining

function PL-FC-ENTAILS?(KB, q) **returns** *true* or *false*

inputs: KB , the knowledge base, a set of propositional definite clauses

q , the query, a proposition symbol

$count \leftarrow$ a table, where $count[c]$ is the number of symbols in c 's premise

$inferred \leftarrow$ a table, where $inferred[s]$ is initially *false* for all symbols

$agenda \leftarrow$ a queue of symbols, initially symbols known to be true in KB

while $agenda$ is not empty **do**

$p \leftarrow \text{POP}(agenda)$

if $p = q$ **then return** *true*

if $inferred[p] = \text{false}$ **then**

$inferred[p] \leftarrow \text{true}$

for each clause c in KB where p is in $c.PREMISE$ **do**

decrement $count[c]$

if $count[c] = 0$ **then** add $c.CONCLUSION$ to $agenda$

return *false*

Inference – Forward Chaining

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

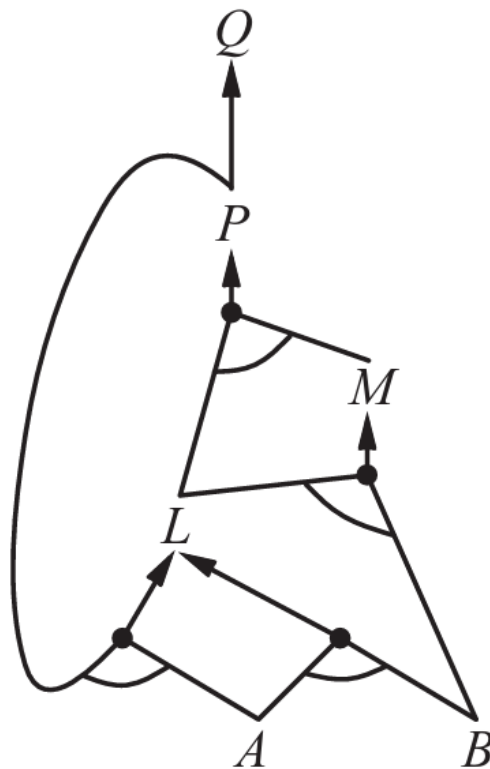
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

A

B



Inference – Forward Chaining

$$P \Rightarrow Q$$

$$L \wedge M \Rightarrow P$$

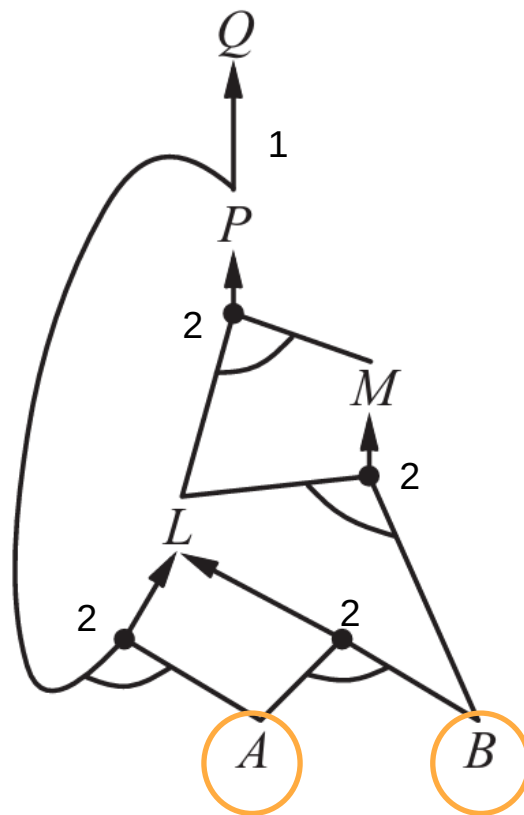
$$B \wedge L \Rightarrow M$$

$$A \wedge P \Rightarrow L$$

$$A \wedge B \Rightarrow L$$

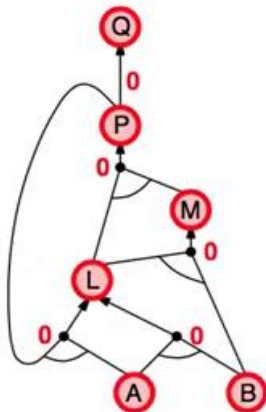
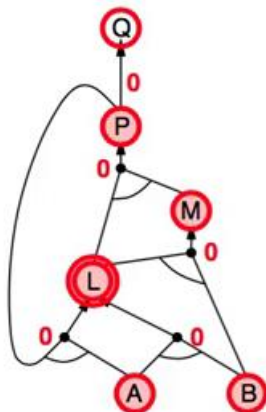
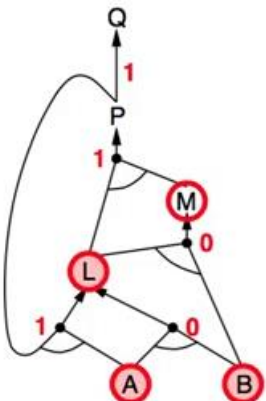
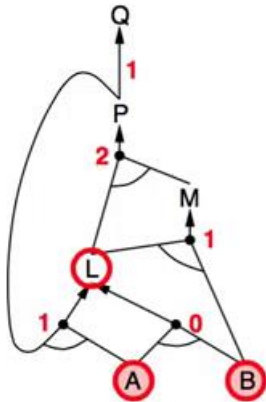
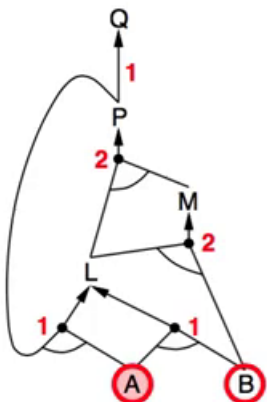
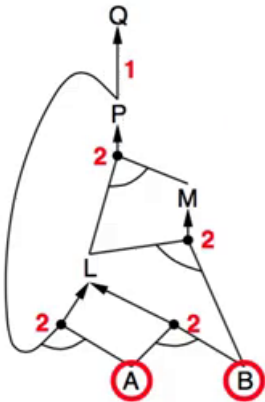
A

B



Inference – Forward Chaining

$P \Rightarrow Q$
 $L \wedge M \Rightarrow P$
 $B \wedge L \Rightarrow M$
 $A \wedge P \Rightarrow L$
 $A \wedge B \Rightarrow L$
 A
 B



Inference – Backward Chaining

Works backwards from query q

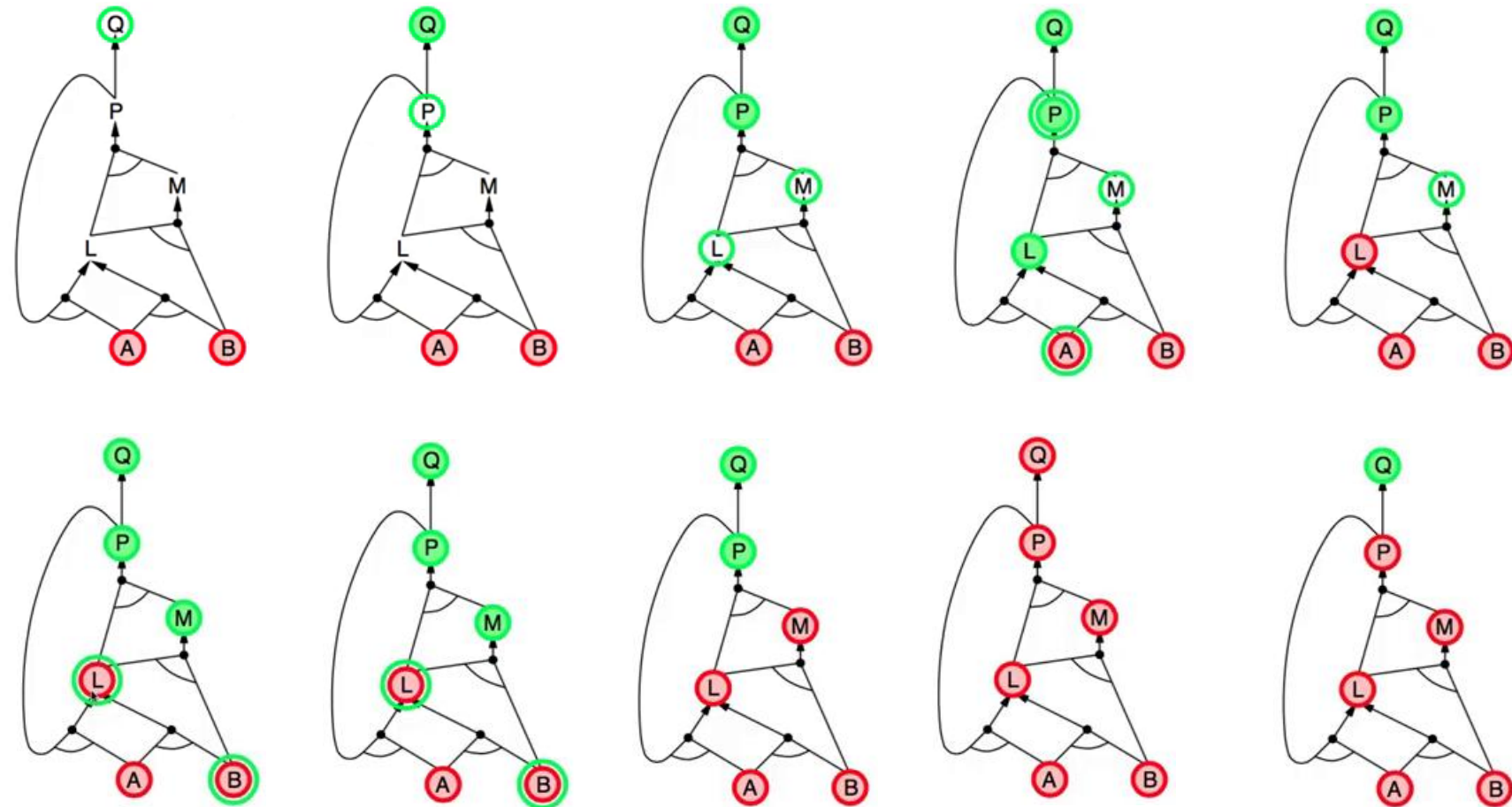
- To prove q by B.C.
- Check if q is known or
- Prove by B.C. all premises of some rule concluding q

Avoid Loops: Check if new subgoal is already in goal stack

Avoid repeat work: Check if new subgoal

- Has already been proved true or
- Has already failed

Backward Chaining



Forward and Backward Chaining - Discussion

- FC is data driven. E.g. object recognition, routine decision
- FC may do a lot of work irrelevant to the goal
- BC is goal driven. Appropriate for problem solving. i.e.
- Where is home? What's the result of equation x ?
- Complexity of BC can be much less than linear size of KB

Thank You

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